MUSTAFA YAMIN

AI Engineer | FastAPI | Agentic AI | RAG | Generative AI | Full-Stack Development

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience designing and deploying agentic AI systems, retrieval-augmented generation (RAG) pipelines, and computer vision solutions in production environments. Skilled in Python, FastAPI, and the Azure AI ecosystem, with a track record of leading client-facing projects end-to-end - from architecture through deployment. Combines strong technical execution with team leadership and client coordination experience. BSCS candidate at Virtual University of Pakistan.

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, TypeScript/JavaScript, Next.js, React, TailwindCSS, Shadcn UI

AI / ML: Agentic AI, Claude Agent SDK, OpenAI Agents SDK, Prompt Engineering, Retrieval-Augmented Generation (RAG), GPT-4o, OpenAI Embeddings, YOLO / Computer Vision, OpenCV

Cloud & DevOps: Azure AI Foundry, Azure Bot Service, Azure Cosmos DB, Azure Fabric, Azure Blob Storage, Azure Speech (STT/TTS), Azure AI Search, Azure App Service, Azure Authentication, Docker, Ubuntu Server (SSH) Deployment

Data & Automation: PostgreSQL, Medallion Architecture, Power BI, Power Automate, Odoo, FFmpeg

Other: SQL & NoSQL Databases, Keyword Research & SEO, Content Writing, Social Media Management

PROFESSIONAL EXPERIENCE

AI Engineer | Inseyab Consulting & Information Solutions LLC

April 2026 - Present | NASTP, Karachi, Pakistan

- Led development, as senior developer, of a client-facing sales executive AI agent supporting both text and voice conversations, coordinating directly with the client and their technical team on requirements, architecture, and delivery.
- Built a RAG pipeline using Azure AI Search and OpenAI's text-embedding-3-small, integrated with GPT-4o via Azure AI Foundry, to power accurate, context-aware sales conversations.
- Implemented voice interaction using Azure Speech Services (STT/TTS) and an automated pipeline that generates a daily audio podcast summary, storing output in Azure Blob Storage.
- Automated daily sales summary emails via Power Automate and built on-demand Power BI chart generation driven by natural-language user requests at runtime.
- Designed a Medallion Architecture data pipeline on Azure Fabric for structured, governed data storage, with session state managed in Azure Cosmos DB and secure access enforced via Azure Authentication; built the API layer in Python (FastAPI) and deployed the application on Azure App Service.
- Integrated a YOLO computer vision model into a warehouse dispatch automation system, processing live camera streams frame-by-frame with OpenCV to detect, count, and track sacks loaded onto and off trucks in real time.
- Mapped model-generated bounding boxes and segmented coordinates onto the live video feed to compute accurate real-time counts, using FFmpeg for video streaming and FastAPI as the middleware layer between two PostgreSQL databases (custom middleware DB and Odoo's warehouse management DB).
- Implemented Next.js frontend to surface live counts and dispatch status, containerized the stack with Docker, and deployed to a production Ubuntu server via SSH.
- Deployed and maintained multiple production applications on Ubuntu servers, ensuring reliable Docker-based and Azure-hosted delivery across projects.

Junior Full Stack Developer | E-services and Technologies

October 2025 - April 2026 | Pakistan

- Built responsive web applications and dynamic user interfaces integrated with Generative and Agentic AI, working across the full stack with React, Next.js, Angular, and the OpenAI Agents SDK.
- Developed front-end and back-end features using HTML, CSS, JavaScript/TypeScript, TailwindCSS, Node.js, and both SQL and NoSQL databases, focused on clean UI/UX and performance.
- Collaborated in cross-functional teams to solve technical challenges and continuously adopted modern frameworks, cloud platforms, and AI integrations.

Web Development Intern | E-services and Technologies
February 2025 - October 2025 | Karachi, Pakistan

- Supported web development projects, building front-end components and gaining hands-on experience with modern JavaScript frameworks ahead of promotion to Junior Full Stack Developer.

AI and Aerospace Coordinator / Stream Coordinator | Stream Academy International
May 2024 - Present | Karachi, Pakistan

- Organize, research, and implement AI solutions while leading engaging, hands-on STEM sessions and workshops for students.
- Guide students on Web Development and AI projects, as well as aerospace concepts including High Altitude Balloons and Cube Satellites.
- Foster creative and critical thinking through project-based learning; track student progress and prepare reports on learning outcomes.

Content Research & Writer | Engagex Digital
March 2024 - September 2024 | United States (Remote)

- Researched topics from reliable sources and conducted keyword research to identify content optimization opportunities.
- Wrote SEO-optimized, engaging blogs, articles, and product reviews; proofread and edited content for accuracy and clarity.
- Managed content publishing and distribution across multiple platforms.

Volunteer Challenge Master | Destination Imagination Pakistan
2023 | Karachi, Pakistan

- Evaluated and appraised team performances at Destination Imagination's Pakistan National Tournament, fostering creativity and problem-solving in student teams.

PROJECTS

- AI Event Planner
- E-commerce Marketplace - the-brands-bazar.vercel.app
- Knowledge-Hub (Single-Page Blog Platform) - knowledge-hub-ebon.vercel.app
- Static E-commerce UI - ecommerce-static-website.vercel.app
- Resume Builder - static-resume-builder-ruddy.vercel.app
- Tic-Tac-Toe Game - tic-tac-toe-nine-xi-74.vercel.app
- Countdown Timer App - countdown-timer-one-brown.vercel.app

EDUCATION

Bachelor of Science, Computer Science - Virtual University of Pakistan
March 2025 - December 2028 (In Progress)

Generative AI, Web3 & Metaverse Program - Governor Sindh Initiative
Completed February 2024

CERTIFICATIONS

- Building with the Claude API
- Claude Code in Action
- Introduction to Agent Skills
- IBM Dev Day: AI Demystified
- International Space Apps Challenge